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DEPARTMENT OF INFORMATION AND COMPUTER
ENGINEERING

MANAGEMENT OF CINEMA TICKET BOOKINGS
PART 2
SOFTWARE REQUIREMENTS SPECIFICATION

STUDENT DETAILS

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1.0. Introduction

1.1. Purpose

The purpose of the document is to present a detailed description of the Cinema Ticket Reservation Management system. It will clarify the purpose, characteristics, interfaces and interaction of the system with an external entity, as well as the functions and constraints on which the system must operate.

1.2. Scope of the Project

The software system will be a Movie Ticket Reservation Management system for a movie theater that wants to handle the ticket status of all the movies it has available to customers. Behind the system is also a relational database, which contains a list of customers registered in the system (user password, access code, full name, date of birth, residential address, mobile phone, email , active reservations), a list of available movies (movie name, screening room, screening type, screening time, number of tickets available, cost of each available ticket), active bookings (movie details, customer details, number of tickets, ticket cost), and a list of customers who wish to be informed by immediate notification in case a position becomes vacant.

Additionally, the system is designed so that the cinema has full control over the ticket bookings for the films it has and the customer can secure their ticket to watch the film of their choice. The system also provides interaction with a secondary external observer beyond the primary external observer, i.e. the customer, and that is the banking system managed by the respective bank with which the customer cooperates. The banking system manages the operations related to making the transaction for the purchase of tickets.

1.3. Operating Environment

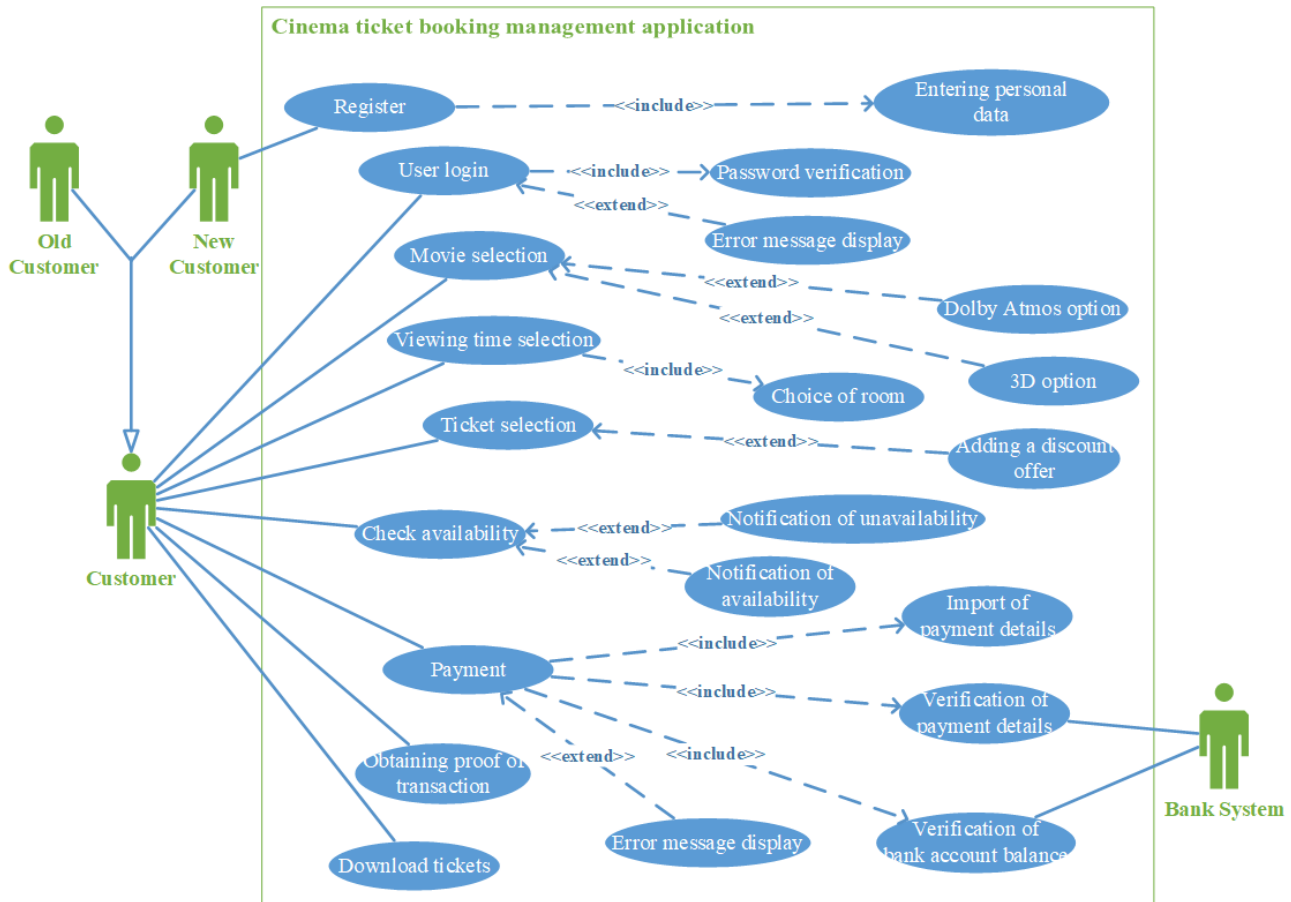


Figure 1 - Movie Ticket Reservation Management System

The Movie Ticket Reservation Management system has two external observers, the Customer and the Bank system. The Customer is the main observer that interacts with the system directly, while the Bank System is a secondary observer that only interacts if the main observer (Customer) interacts with the system. Furthermore, the 'Customer' entity is specialized into 'New Customer' and 'Old Customer' sub-entities, as the first function that the 'New Customer' sub-entity interacts with is the 'Register' function where it is only necessary for entity's first interaction with the system. After "Registration" the functions are common to both sub-entities and therefore the associations are made to the general entity "Customer".

1.4. Cases and Dependencies

List any hypothetical factor that could affect the system requirements. Examples of hypothetical factors could include third-party or commercial components that you plan to use, problems with your development or operating environment, or limitations. The project may be affected if these assumptions are in error, not shared, or changed. Additionally, identify any dependencies the project has on external factors, such as software components that you wish to reuse from another project, unless they are already documented elsewhere.

1.5. Document Overview

The next chapter, Requirements Specifications of this document is primarily written for developers and describes in technical terms the details of the product's functionality.

2.0. Requirements Specifications

2.1. External Interface Requirements

A connection to an external system is required, with the bank's system verifying payment details and transaction status (checking the customer's bank account balance). The Bank's system is activated when the customer selects the "Payment" use case, in order to verify the payment and transaction details. In addition, from the relational database of the application system, the fields recorded in the Customer Status Database are the customer's first and last name, mobile phone, email, user password and access code. The system achieves fast service by keeping the details of registered customers in the database, so that they can easily and quickly assign their details for one or more reservations. Also, the database keeps information about the status of the tickets with the fields movie name, screening room, type of screening, screening time, number of available tickets, cost of each available ticket. Finally, the database also holds information on active reservations with fields such as movie details, customer details, ticket number, ticket cost). If the availability of tickets is zero, then after the consent of the customer himself, his details are entered in the list for immediate notification in case even one seat becomes free.

2.2. Functional Requirements

2.2.1. Recording

Use Case Name	Recording
Inception	The Customer chooses to register in the system
Condition	The Customer is not registered in the system database (Sub-Entity New Customer)
Basic Path	1. The Customer enters his personal information 2. The Customer enters a username and password
Alternative Paths	In Step 2, if the Customer enters a password or code that does not contain an uppercase letter or number 1. An error message appears for registration failed In Step 2, if the Customer enters a password or code that contains an uppercase letter or number 1. A confirmation message appears for successful registration
Postcondition	The Customer's details are entered into the system's database
Exception Paths	The Customer has not filled in a required field

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2.2. 2. User Login

Use Case Name	User login
Inception	The Customer chooses to log into the system.
Condition	The Customer is registered in the database (Sub-Entity Old Customer)
Basic Path	1. The Customer enters his password 2. The Customer enters the password 3. The database verifies the Customer's login details in the system
Alternative Paths	In Step 3, if the input data is not verified by the database 1. An error message appears In Step 3, if the input is verified by the database 1. A message appears for successful verification and login of the user to the system
Postcondition	The Customer has access to the system
Exception Paths	The Customer enters login information that is not verified by the database or tries to log in without completing a field

2.2.3. Movie selection

Use Case Name	Movie selection
Inception	The Customer chooses the movie he wants to watch
Condition	The film to be screened by the cinema
Basic Path	1. The Customer chooses the movie he wants to see 2. If there is, he can choose to watch it in Dolby Atmos for a better sound experience 3. If there is, he can choose to see it in 3D
Alternative Paths	In Step 2 and 3, if there is Dolby Atmos or 3D version of the film respectively and the Customer chooses to see one of these versions 1. It moves on to the next process 2. Otherwise it chooses to see the simple version
Postcondition	The Customer has access to the viewing times
Exception Paths	If the available films are not to the Customer's liking, he may abandon the process

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2.2. 4. Select a viewing time

Use Case Name	Select a viewing time
Inception	The Customer chooses a viewing time
Condition	Have viewing time available
Basic Path	1. The Customer chooses a viewing time 2. The Customer chooses the screening room
Alternative Paths	In Step 1 the available viewing times may not serve the Customer 1. Proceeds to search for another movie 2. Otherwise, it abandons the process
Postcondition	The Customer has access to the available seats in the hall
Exception Paths	If the viewing hours do not suit the Customer, he may abandon the process

2.2.5. Selection of tickets

Use Case Name	Selection of tickets
Inception	The Customer selects a number of tickets
Condition	Availability of tickets
Basic Path	1. The Customer chooses the number of tickets with a limit of 9 tickets 2. Adds, if eligible, discount offer
Alternative Paths	In Step 2 if the customer does not meet the criteria of the discount offer 1. The discount is rejected 2. Otherwise it is validated
Postcondition	The Customer's request is sent to the system
Exception Paths	The Customer may exceed the maximum number of tickets he can book

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2.2. 6. Availability check

Use Case Name	Check availability
Inception	The Customer chooses to search for tickets
Condition	The Customer wishes to search from 1-9 tickets
Basic Path	1. The Customer chooses to search for the desired tickets 2. A check is made for availability and whether the system can respond to the customer's request
Alternative Paths	In Step 2 if the number of tickets requested corresponds to availability 1. The selected positions are held for 8 minutes in the customer's possession in order to complete the purchase, otherwise they are again made available for purchase to other interested parties In Step 2 if the free seats are less than the desired number of seats to reserve 1. The Customer is asked if he wishes to continue the process with these positions 2. The Customer answers positively or negatively In Step 2 if there is no availability at all 1. The Customer is asked if he wishes to be on the list for immediate notification in the event that even one seat becomes available 2. The Customer answers positively or negatively
Postcondition	The Customer is informed about the availability of the tickets
Exception Paths	In case of lack of availability, the Customer can leave the process directly, before being asked if he wishes to be added to the list for immediate notification of any vacancy

2.2.7. Payment

Use Case Name	Payment
Inception	The Customer chooses to pay his reservation
Condition	The Customer must complete the payment information
Basic Path	1. The Customer chooses to pay his reservation 2. Enter payment details 3. Payment details are verified by the cooperating banking system 4. His bank account balance is verified 5. As long as there is no obstacle, the transaction goes through
Alternative Paths	In Step 3, if the payment details are not verified by the banking system 1. A transaction failed message appears 2. Otherwise a successful verification message is displayed In Step 4, if the customer's bank account balance is insufficient for the transaction 1. A transaction failed message appears 2. Otherwise a successful transaction message is displayed
Postcondition	An active booking once successfully completed with payment has been saved in the system database
Exception Paths	When entering the payment details, the Customer can leave a mandatory field blank.

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2.2. 8. Receive proof of transaction

Use Case Name	Receive proof of transaction
Inception	The banking system successfully verifies the customer's payment details and account balance
Condition	The transaction is completed successfully
Basic Path	1. The banking system gives approval for the successful completion of the transaction 2. The system merges into one document, the transaction receipt 3. The system sends the receipt to the customer
Alternative Paths	None
Postcondition	The transaction proof is available for the Customer to download
Exception Paths	None

2.2.9. Get tickets

Use Case Name	Get tickets
Inception	The system has sent the Customer the transaction receipt
Condition	The transaction is completed successfully
Basic Path	1. The system integrates into the necessary documents, the tickets with the necessary information for their certification by the cinema employees 2. The system sends the tickets to the Customer
Alternative Paths	None
Postcondition	Tickets are available for the Customer to download
Exception Paths	None

2.3. Detailed Non-Functional Requirements

2.3.1. Safety

The SQL server which handles the system database, has its own security to prevent untrusted write / delete accesses . There is no restriction on read access .

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Thank you for your attention.

